BENCHMARKING RECURRENCE CFD with SCALAR TRANSPORT IN A 3D LID-DRIVEN CAVITY FLOW

Md Zahid Hasan

Supervisor: Professor Dr. Stefan Pirker

JOHANNES KEPLER UNIVERSITY LINZ

Motivation & Core Idea

The Problem
Computational Cost
Traditional CFD re-solves the full **Navier–Stokes** system for *every* new scalar configuration, taking hours or days per run.

The Insight
Turbulence Repeats
Steady turbulent flows revisit states on a bounded **attractor**. rCFD identifies and stores these recurring pathways *once*.

The Solution
Hours → Seconds
Scalars are advanced directly along pre-mapped paths with **no repeated flow solving**, enabling massive parametric studies instantly.

CFD vs Recurrence CFD

Turbulent Flow	Re-solve	Hours / Days	Capture Once	Advance	Seconds
CFD (N-S Solve)	per scalar	Per config	Recurring paths	Scalars along path	Any config

$\sim 10^4\times$ **Speed-up factor** vs. full N-S re-solve per scalar

$1\times$ **Flow solve needed** Reused for all scalar configurations

∞ **Parametric studies** Enabled at near-zero additional cost

Core Principle – rCFD

"rCFD leverages the statistical recurrence of quasi-periodic, statistically-stationary turbulent flows — once the flow's recurrence structure is characterized via the recurrence matrix, passive-scalar transport for any inlet configuration is reduced to a lightweight sequence-stitching and cell-to-cell shift operation, entirely bypassing repeated Navier–Stokes solves and enabling parametric studies at negligible computational cost."

Introduction

The lid-driven cavity is one of the most extensively studied benchmark configurations in computational fluid dynamics, providing a well-defined recirculating flow field that serves as an ideal controlled environment for method validation. This work benchmarks the Recurrence CFD (rCFD) methodology against full Navier–Stokes simulations for passive scalar transport in a three-dimensional lid-driven cavity, with nitrogen as the working fluid at $\mathrm{Re} \approx 1000$.

The complete workflow encompasses three phases:

Spin-up run Establish stationary baseline flow → **Database** Extract & store recurrence → **rCFD Prediction** Predicts mixing in $\leq$ **10 s**

To evaluate rCFD's predictive capability, a passive scalar is injected into the top-right sector of the cavity and tracked over 20 s under two contrasting diffusivity regimes. The results demonstrate that rCFD accurately reproduces both the **mixing index** and the **centre-of-gravity** trajectories obtained from high-fidelity CFD – correctly capturing the dominant vortex-driven redistribution and diffusive homogenisation – within seconds. Also, contours of the mixing are recorded per time-step.

Working Mechanism of rCFD

PHASE I : GOAL

*Sample the **stationary attractor** — the set of recurring flow patterns the turbulence revisits — so that all future scalar transport runs can reuse these paths without re-solving the Navier-Stokes equations.*

High-Fidelity CFD Run —solve N–S→ **Data Extraction** —map R_{ij}→ **Stationary Attractor Sampled**

CFD solver captures fast unsteady dynamics over the full spin-up window

Convection & diffusion fields sampled every 5 solver steps → 200 snapshots

Recurrence matrix R_{ij} built; stitching path identified

CONVECTION

Cell-to-Cell Shift Vectors

Pre-computed **cell-to-cell shift vectors**, extracted from stored velocity field snapshots, encode how fluid parcels are displaced across the mesh, replacing convective Navier–Stokes solves along the stationary flow attractor.

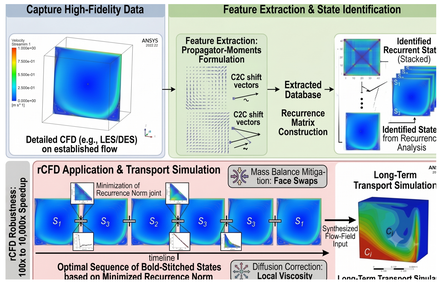
DIFFUSION

Face-Swap Operations

Diffusion is replaced by **face swaps** — scalar concentration is exchanged between neighbouring cells across shared faces, with the swap frequency governed by local sub-grid viscosity to reproduce physically consistent turbulent mixing.

Working mechanism of rCFD

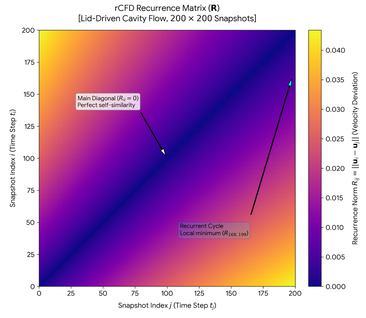
Phase II – The rCFD Acceleration Engine



Core Workflow:

- **1. Extract Data:** Run high-fidelity simulation to sample attractor.
- **2. Map Attractor:** Compute Recurrence Matrix (R).
- **3. Stitch Sequence:** Use R to find smooth flow transitions.
- **4. Transport Solve:** Decouple fluid **u** from passive scalars c_i.

Recurrence Matrix Anatomy



Pairwise Recurrence Matrix **R** (200 × 200)

1. Mathematical Formulation

Each entry (i, j) of **R** is the L_2-norm velocity deviation between snapshots t_i and t_j:

$$R_{ij} = \|\mathbf{u}_i - \mathbf{u}_j\|_2$$

Low R_{ij} marks mutually similar frames — the prerequisite for valid sequence stitching.

2. Key Structural Properties

- **Main Diagonal ($R_{ii} = 0$):** Zero norm of perfect self-similarity.
- **Symmetry ($R_{ij} = R_{ji}$):** L_2 is a proper distance metric.
- **Recurrence Prominence:** Depth of the off-diagonal minimum relative to the surrounding plateau; low prominence causes arbitrary frame pairings and unphysical transitions.
- **Cycle Closure ($R_{168,199} \approx 0.0051$):** Near-identical snapshots confirm the flow revisits a prior state, enabling a discontinuity-free synthetic loop.
- **Attractor Centroid (t_{100}):** Minimises the row-mean $\bar{R}_{100} \approx 0.0108$ — statistically closest to all other flow states.

Phase III – Stitching

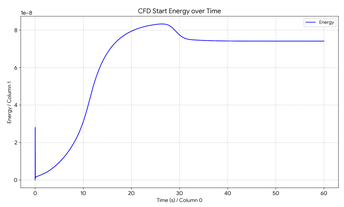
1. **Recurrence Identification:** Compute the pairwise recurrence matrix **R** from stored velocity snapshots; identify mutually similar frame pairs via the recurrence norm $R_{ij} = \|\mathbf{u}_i - \mathbf{u}_j\|_2$.
2. **Flow Reconstruction:** Stitch selected recurrent sequences to compose a continuous, long-term artificial flow field $\mathbf{u}_{\mathrm{rec}}(t)$, bypassing repeated Navier–Stokes solves.

Run rCFD in ANSYS Fluent (Windows)

Three sequential phases, each automated by a Scheme script.

Phase 1 - Spin-Up CFD `CFD_start.scm`

Solves unsteady N–S (SIMPLE, k–ω, Δt=0.01 s) for **60 s** (600 episodes × 10 steps) from `_INIT.cas.h5`. UDF monitors $E_{\mathrm{kin}}(t)$ until plateau ⇒ stationary attractor confirmed. Outputs `_START.cas.h5` — the baseline for all downstream scripts.



Why a Dedicated Start-up Run?

- Initial condition (e.g. quiescent fluid) contains strong transients.
- Spin-up drives mean vortex core, secondary recirculation zones, and k–ω fields.
- Spin-up snapshots introduce spurious recurrence structures and skew ensemble-averaged mixing statistics.

Two main purposes:

1. Generate converged, physically meaningful base flow → `_START` case.
2. Provide kinetic energy history $E_{\mathrm{kin}}(t)$ to identify statistically stationary regime.

"A clean start-up guarantees trustworthy rCFD and scalar transport results."

Phase 2 - Database Prep `rCFD_prep.scm`

Monitored **5 s** run (203 ep, Δt=0.005 s) over `_START`:

p1/p2	Copy `_START` files; compile `libudf`
p3	Allocate 10 UDMI slots; bind coordinates
p4a	Spawn 9 UDS fields (μ_t); calibrate Γ_{eff}
p5	Record scalar/vel snapshots every 5 steps
p7	Compute R_{ij}; identify stitching path

Phase 3 - Prediction `rCFD_run_topright.scm`

Advances scalars along path for **20 s** (800 ep, Δt=0.025 s)

MIX1	`diffusivity` $= 2\times10^{-5}$; slow diffusion
MIX2	`diffusivity` $= 0.03$; rapid homogenise
Output	Mix index → 1.0 over 20 s; Center of Gravity trajectory; contour images

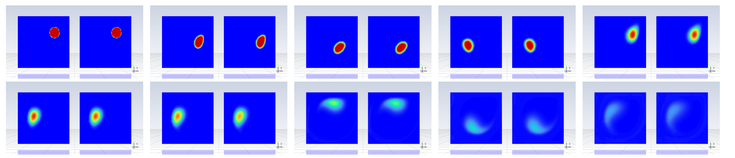
The database is computed *once* from a single full-fidelity CFD run. The recurrence matrix encodes the complete attractor of the statistically-stationary flow as a finite snapshot library. Every subsequent transport query is reduced to a lightweight path traversal through stored flow states. This one-time cost unlocks speedups of $\mathcal{O}(10^2)$–$\mathcal{O}(10^4)\times$ over repeated full CFD solves.

rCFD Prediction Results

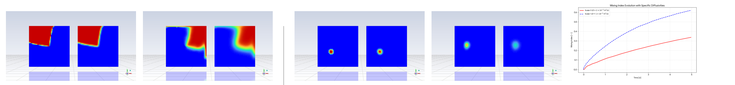
Boundary Conditions	
Density $= 1.7\,\mathrm{kg/m^3}$	Viscosity $= 1.76\times10^{-5}\,\mathrm{kg/(m\cdot s)}$
Lid velocity $= 0.30\,\mathrm{m/s}$ (along x)	Turbulence model = k–ω SST

rCFD_user_run_topright.h (config header)	
Spec: Scalar transport, seeding ($\phi = 1$)	**Target:** Homogenise toward $\phi = 0$
Domain: $0.05 \times 0.05 \times 0.005$ m	Centre: $(C_x, C_y) = (0.035, 0.035)$ m
Radius: $C_R = 0.005$ m	Depth: $z \leq 0.005$ m

Mixing progress tracked via rCFD contour snapshots



Place any shape at any position to predict $\leq$ 10 seconds



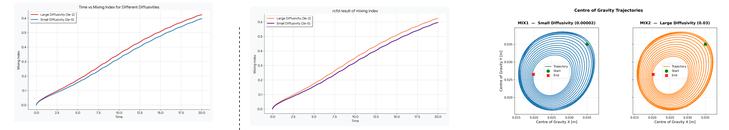
rCFD comparison with CFD Simulation

Mixing Index (Final)

$$MI(t) = 1 - \frac{\sum V_c \left|\phi_c - \bar{\phi}\right|}{\sum V_c \left|\phi_{c,0} - \bar{\phi}_0\right|}$$

Center of Gravity (Final)

$$x_{\mathrm{CoG}}(t) = \frac{\sum \phi_c x_c V_c}{\sum \phi_c V_c}, \qquad y_{\mathrm{CoG}}(t) = \frac{\sum \phi_c y_c V_c}{\sum \phi_c V_c}$$



Reference CFD | **rCFD Results**

rCFD: Performance Advantages & Limitations

Scalability & Robustness

✓ **One-Time Investment**
Run N–S once; reuse DB for all scalars.

✓ **Lightweight Iteration**
Update `rCFD_user.h` → predictions in **seconds**.

Computational Cost and Time reduction

Known Limitations

× **Strong Transients**
Fails if attractor not yet formed (unsteady start-up).

× **Chaotic Turbulence**
No recurring patterns no usable stitching path in R_{ij}.

× **Not for Changing geometry or fluid rheology**

Future Work

- **GPU Acceleration:** proof of concept to transfer the pipeline CPU to GPU.
- Compare results against **Physics-Informed DeepONet (PI-DeepONet)** on identical simulation.

References

1. Lumetzberger, H., Pirker, S., & Lichtenegger, T. (2025). Propagator-moments approximation for recurrence CFD: Application to species transport in turbulent flows. *Chemical Engineering Science, 311*, 121624. https://doi.org/10.1016/j.ces.2025.121624